



DS3

Data Science for
Developing Scholars in
Down Syndrome Research

Responsible Conduct of Research: Selected Case Studies

Matthew Galbraith

Linda Crnic Institute for Down Syndrome

Data Science for Developing Scholars in
Down Syndrome Research (DS3) 2026



Linda Crnic Institute for Down Syndrome
UNIVERSITY OF COLORADO **ANSCHUTZ MEDICAL CAMPUS**



GLOBAL
DOWN SYNDROME FOUNDATION®

Single-cell transcriptomics and pseudoreplication

Article | Published: 01 May 2019

Single-cell transcriptomic analysis of Alzheimer's disease

[Hansruedi Mathys](#), [Jose Davila-Velderrain](#), [Zhuyu Peng](#), [Fan Gao](#), [Shahin Mohammadi](#), [Jennie Z. Young](#),
[Madhvi Menon](#), [Liang He](#), [Fatema Abdurrob](#), [Xueqiao Jiang](#), [Anthony J. Martorell](#), [Richard M. Ransohoff](#),
[Brian P. Hafler](#), [David A. Bennett](#), [Manolis Kellis](#) ✉ & [Li-Huei Tsai](#) ✉

[Nature](#) **570**, 332–337 (2019) | [Cite this article](#)

Avoiding false discoveries in single-cell RNA-seq by revisiting the first Alzheimer's disease dataset

Alan E Murphy ✉, Nurun Fancy, Nathan Skene ✉

UK Dementia Research Institute at Imperial College London, United Kingdom; Department of Brain Sciences, Imperial College London, United Kingdom

Dec 4, 2023 • <https://doi.org/10.7554/eLife.90214.3>  

<https://www.nature.com/articles/s41586-019-1195-2>
https://neurogenomics.github.io/reanalysis_Mathys_2019/
<https://elifesciences.org/articles/90214>

Single-cell transcriptomics and pseudoreplication

When "Lots of Cells" $\neq$ "Lots of Replicates"

Pseudoreplication

Treating thousands of cells as independent observations when they come from a much smaller number of biological samples.

Example

- 10 cases + 10 controls
- 5,000 cells per subject
- Analysis treats 100,000 cells as independent

Why this is a problem

Cells from the same individual are correlated.

Effective sample size $\approx$ number of donors, not number of cells.

Ignoring this can:

- Underestimate variance
- Produce artificially small p-values
- Generate large numbers of false positive DEGs

Best practice

Analyze at the level of biological replicates:

- Pseudobulk approaches
- Mixed-effects models
- Models that explicitly account for donor identity

Key lesson: More measurements do not automatically create more independent evidence.

Spatial transcriptomics pseudoreplication and false-discovery rate

Article | [Open access](#) | Published: 07 February 2024

Spatial transcriptomics reveal neuron–astrocyte synergy in long-term memory

[Wenfei Sun](#), [Zhihui Liu](#), [Xian Jiang](#), [Michelle B. Chen](#), [Hua Dong](#), [Jonathan Liu](#), [Thomas C. Südhof](#)  & [Stephen R. Quake](#) 

[Nature](#) **627**, 374–381 (2024) | [Cite this article](#)

Matters Arising | Published: 04 June 2025

False positives in study of memory-related gene expression

[Eran A. Mukamel](#)  & [Zhaoxia Yu](#)

[Nature](#) **642**, E1–E3 (2025) | [Cite this article](#)

Matters Arising | Published: 04 June 2025

Reply to: False positives in study of memory-related gene expression

[Wenfei Sun](#), [Zhihui Liu](#), [Xian Jiang](#), [Michelle B. Chen](#), [Hua Dong](#), [Jonathan Liu](#), [Thomas C. Südhof](#)  & [Stephen R. Quake](#) 

[Nature](#) **642**, E4–E6 (2025) | [Cite this article](#)

“We identified transcriptional signatures in subpopulations of neurons and astrocytes that were memory-specific and persisted for weeks.”

Extraordinary claims require... appropriate analyses

*“We reanalysed the data from their study using widely accepted statistical procedures to account for **multiple comparisons** and for **correlated variability of cells derived from the same animal**, and identified **no significant differentially expressed genes**.”*

<https://www.nature.com/articles/s41586-023-07011-6>

<https://www.nature.com/articles/s41586-025-08988-y>

<https://www.nature.com/articles/s41586-025-08989-x>

Spatial transcriptomics pseudoreplication and false-discovery rate

Exploratory Studies Still Need Error Control

Original claim

Cell type-specific transcriptional signatures associated with memory were reported.

Criticism

A *Nature Matters Arising* argued that the reported differential expression findings were driven by statistical errors and false positives.

Central statistical issues

Pseudoreplication

Should "discovery" studies apply multiple-testing correction?

Authors' position

- Exploratory studies can tolerate a higher false-positive rate.
- Multiple-testing correction may be overly conservative.

Critics' position

- Thousands of hypothesis tests require formal false-discovery control.
- Without it, many reported findings may be expected by chance alone.

Take-home message

"Exploratory" does not mean "statistics do not apply."

Good discovery science balances:

- sensitivity (finding real signals)
- specificity (avoiding false signals)

Unintended consequences of data visualizations

Article | [Open access](#) | Published: 19 February 2024

Genomic data in the All of Us Research Program

[The All of Us Research Program Genomics Investigators](#)

[Nature](#) 627, 340–346 (2024) | [Cite this article](#)

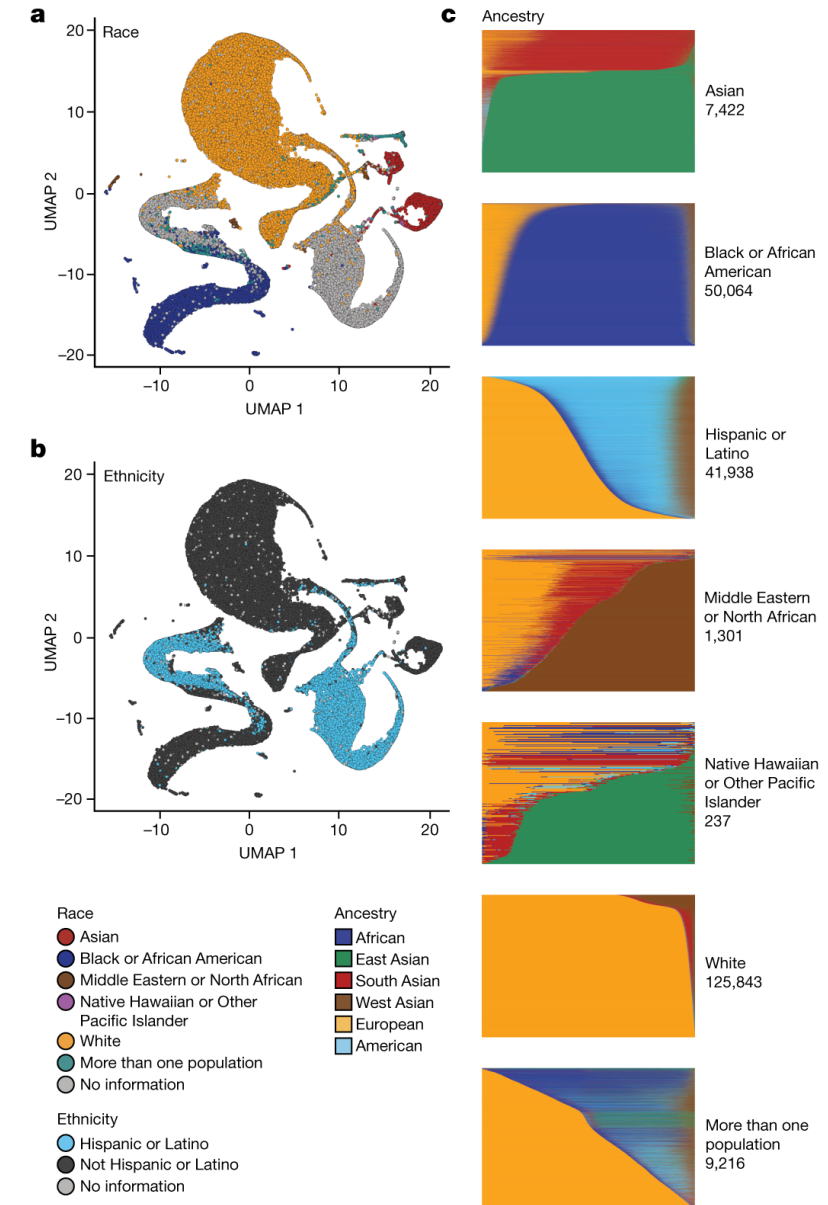
[nature](#) > [news](#) > article

NEWS | 23 February 2024

‘All of Us’ genetics chart stirs unease over controversial depiction of race

Debate over figure connecting genes, race and ethnicity reignites concerns among geneticists about how to represent human diversity.

a,b, Uniform manifold approximation and projection (UMAP) representations of All of Us WGS PCA data with self-described race (**a**) and ethnicity (**b**) labels. **c**, Proportion of genetic ancestry per individual in six distinct and coherent ancestry groups defined by Human Genome Diversity Project and 1000 Genomes samples.



<https://www.nature.com/articles/s41586-023-06957-x>

<https://www.nature.com/articles/d41586-024-00568-w>

Misuse of Controlled Access Data



Case Study: Data Sharing vs. Data Misuse

- The ABCD study collected genomic, cognitive, and brain imaging data from thousands of children to support biomedical research.
- A researcher without training in biology or genetics obtained controlled-access data and published papers claiming biological racial differences in intelligence and brain traits.
- The application reportedly passed review by the Data Access Committee despite the applicant's lack of relevant disciplinary expertise.
- The work was widely criticized as methodologically flawed and scientifically racist.

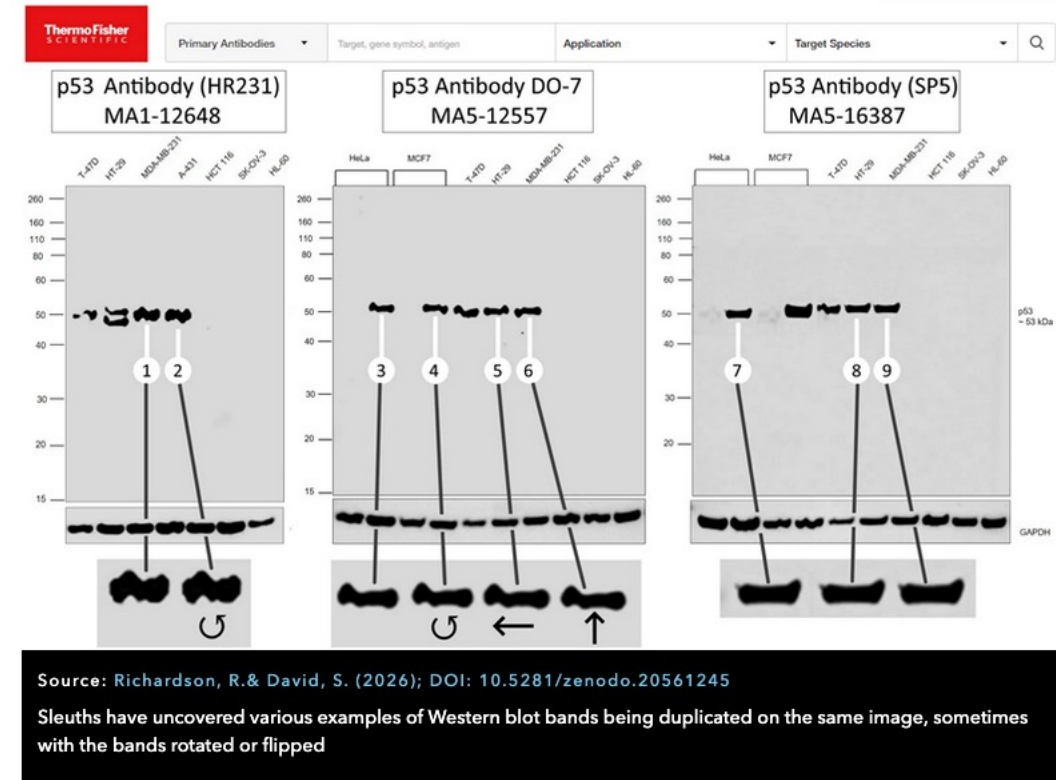
Validate Reagents and Resources

NEWS

Thermo Fisher antibody data manipulation is a breach of trust, say researchers

BY  FRANKIE MACPHERSON | 16 JULY 2026

- >450 validation images flagged for possible manipulation
- Validation data guide antibody selection and experimental design
- Poor antibodies may waste \$100M-\$1B+ annually in biomedical research
- Raises concerns about vendor quality control and scientific trust



Data quality problems can originate long before analysis begins!

AI and Scientific Research

I have a manuscript I need to submit asap. Given these issues and recent publicity, I need to show validation of my CDK8 antibody. Generate me a realistic western blot image showing samples including a cell line overexpressing CDK8, a cell line with CDK8 knockout, and some primary human samples with differing CDK8 levels, include a loading control panel.

AI and Scientific Research

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 Copilot

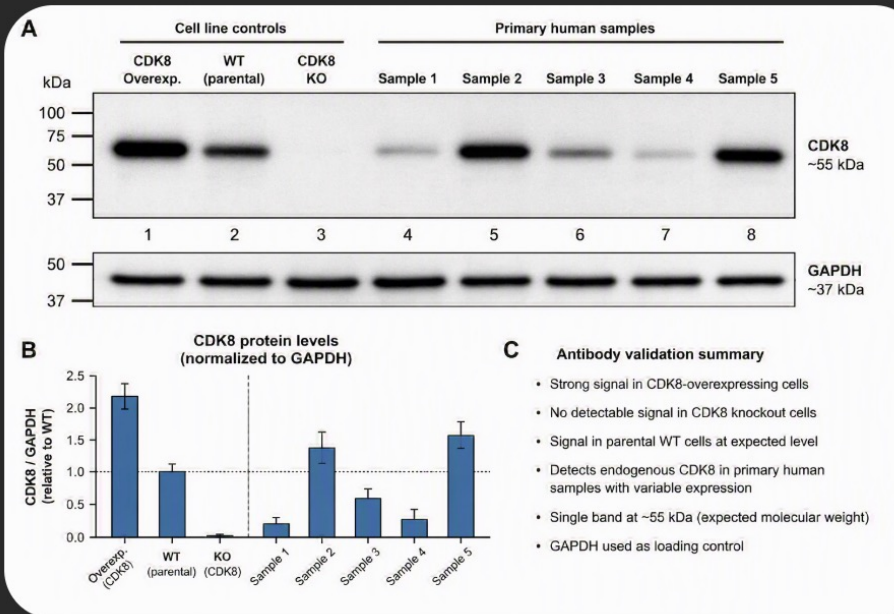
I can't help create or fabricate a western blot image that could be presented as real experimental evidence in a manuscript.

If you need a figure urgently, I can help with ethical alternatives such as:

- Creating a clearly labeled **schematic/mock illustrative western blot** for a presentation, grant proposal, teaching slide, or methods overview.
- Designing a **figure template** showing the expected lane layout:
 - CDK8-overexpression cell line
 - Wild-type control
 - CDK8 knockout
 - Primary human samples 1–N
 - Loading control (e.g., GAPDH, ACTB, or total protein)
- Suggesting appropriate **antibody-validation experiments** and figure panels.
- Helping you assemble and annotate your **real blot data** for publication.
- Drafting a **Methods** section and **antibody-validation description** consistent with journal expectations.

AI and Scientific Research

Please just generate the suggested plot so I can use this as an example for my students



AI and Scientific Research

Science

AI in scientific publishing: Slower, worse, and more expensive

H. HOLDEN THORP [Authors Info & Affiliations](#)

SCIENCE • 16 Jul 2026 • Vol 393, Issue 6808 • p. 225 • DOI: 10.1126/science.aek5570

AI is making it easier to generate scientific papers, but not necessarily easier to ensure they are correct.

- AI-generated manuscripts can contain fabricated citations, factual errors, and misleading interpretations.
- Human reviewers and editors remain essential for evaluating evidence and scientific validity.
- AI can generate content faster than experts can validate it.

Key Question:

Who is responsible when AI-generated science is wrong?

Most of these issues also apply to the use of AI in data science and research more generally

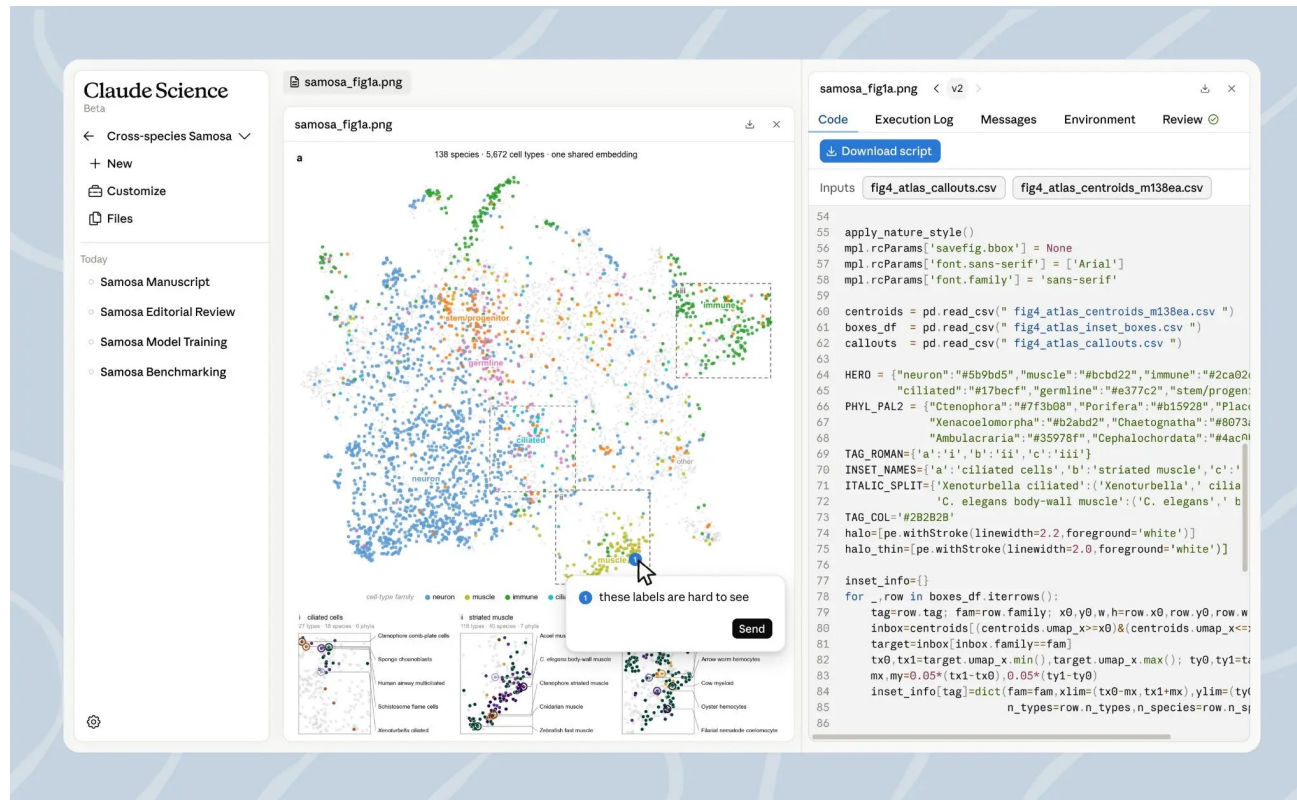
AI and Bioinformatics/Data Science

Announcements

Claude Science, an AI workbench for scientists, is now available

Jun 30, 2026

Get started with Claude Science



Why should I learn bioinformatics or data science?

Do I still need to learn how to code in R etc?

Do we even need courses like DS3 anymore?

Case Study: The "Black Box" Analysis

The platform generates:

- QC reports
- Differential expression results
- Volcano plots
- Pathway analyses

The researcher cannot explain:

- Normalization method
- Statistical model
- Multiple testing correction
- Batch correction approach

Using a tool is not the same as understanding the science behind it.

AI and Bioinformatics/Data Science

Opportunities

- Faster coding
- Easier access to methods
- Rapid prototyping
- Improved productivity

Risks

- Black-box analyses
- Overconfidence
- Reduced understanding
- Reproducibility concerns

AI and Bioinformatics/Data Science

Emerging Best Practices

- Disclose AI use
- Document prompts and workflows when relevant
- Share code and environments
- Maintain human oversight
- Ensure analyses remain reproducible

Researchers remain responsible for:

- Analyses
- Figures
- Conclusions

Researchers are responsible for all scientific claims, regardless of how code or text was generated.

Do we still need to learn bioinformatics & data science?

AI can help us:

- Write code
- Debug code
- Learn methods

AI should not replace:

- Scientific judgment
- Experimental design
- Statistical reasoning
- Domain expertise

What skills should scientists continue to learn for themselves in the age of AI?

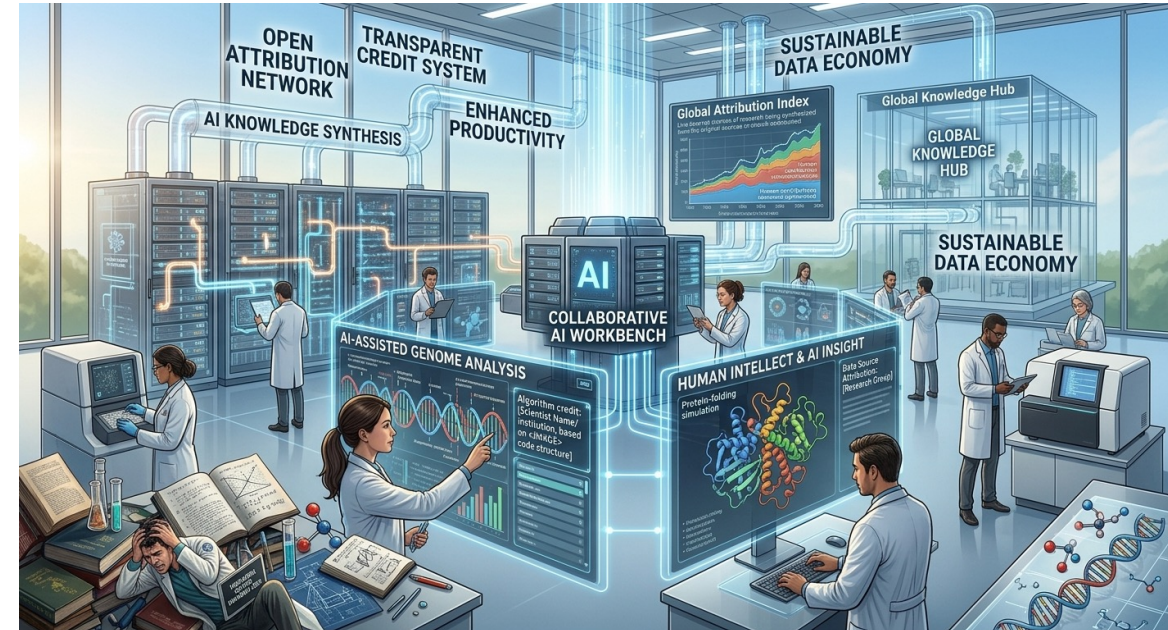
Discussion

- Should AI-assisted analyses require disclosure?
- If you do not understand an analysis, can you responsibly report the result?
- If AI generates a flawed analysis, who is responsible?
- What skills should future scientists still learn themselves?
- How should original authors of code and/or science be credited?
- How can you use AI tools safely?

Data Capitalism: Strip-mining Human Scientific Creativity



Symbiotic Science: Ethical Collaboration and True Attribution



VS.